

D804

# Advanced AI for Computer Scientists

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A wildlife agency's image model already recognizes 200 species. Rangers want it to recognize a newly listed species, but they have only 3 labeled photos of it, taken by a field lab with 40 staff in a mountain region. Which approach fits?

- A. Wait for thousands of photos, since AI always needs large datasets.
- B. Zero-shot learning, since 3 photos are almost none.
- C. Train a new model from scratch on the 3 photos.
- D. Few-shot learning that adapts the existing model using the 3 labeled photos.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

A robotics lab funded by a five-year grant trains a policy across hundreds of different grasping tasks so that the robot can learn to grasp a brand-new object in a few trials. What is this approach?

- A. Supervised learning on one large grasping dataset
- B. Meta-learning
- C. An ensemble of grasping models
- D. Transfer learning from one pretrained model

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A team stacks gradient-boosted trees, a random forest, and logistic regression. It trains the meta-model on predictions the base models made on the same rows they were trained on. Everything runs on a 30-node Kubernetes cluster. What is the problem?

- A. Nothing; stacking always beats each base model.
- B. The cluster is too small for stacking.
- C. The base models should all be the same type.
- D. In-sample predictions are overconfident, so the meta-model learns a false picture; use out-of-fold predictions.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A warehouse robot, made by a firm operating in six countries with an 8-hour battery, will receive a reward of +1 at each of the next three steps, and then the episode ends. The discount factor is 0.9. What is the discounted return from now?

- A. 3
- B. It cannot be found without the 8-hour battery limit.
- C. 2.439
- D. 2.71

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

An agent will receive one reward of 100 exactly two steps from now and nothing before. Its discounted value now is 64. What is the discount factor  $\gamma$ ?

- A. 0.36
- B. 0.64
- C. 0.8
- D. 1.25

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A recommendation agent that serves 3 million users uses  $\epsilon$ -greedy with  $\epsilon = 0.2$  across 4 possible actions; its random choice can land on any action, including the best one. What is the probability it picks the current best action?

- A. 0.25
- B. 0.20
- C. 0.85
- D. 0.80

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7. [TRANSFER — reversed stem / different variable]

An  $\epsilon$ -greedy agent with 5 actions picks its best-known action 92% of the time. What is  $\epsilon$ ?

- A. 0.40
- B. 0.10
- C. 0.08
- D. 0.92

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A fraud detector used by 22 regional banks flags 90% of fraudulent transactions and 5% of legitimate ones. Fraud makes up 1% of transactions. The vendor recently won an industry award. If a transaction is flagged, about how likely is it to be fraud?

- A. About 15%
- B. 1%
- C. 90%
- D. 95%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

9. [TRANSFER — reversed stem / different variable]

A screening model with 99% sensitivity and 99% specificity is applied to a condition that affects 1 in 10,000 people. A person screens positive. Which statement is correct?

- A. The model is useless and should never be used.
- B. The chance the person has the condition is under 1%.
- C. The chance is 99%.
- D. The chance is 50%, since the result is either right or wrong.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

10.

For on-device keyboard prediction, Model A has 94.1% accuracy, is 1.2 GB, and takes 400 ms per prediction; Model B has 93.6% accuracy, is 60 MB, and takes 15 ms. The requirement is under 50 ms on phones. Both were built by a team working across two time zones. Which should be deployed?

- A. Model B — it meets the latency limit at a 0.5-point accuracy cost and uses far less memory and energy.
- B. Neither, since any model under 95% accuracy is unacceptable.
- C. Model A, because it has the highest accuracy.
- D. Model A served from the cloud, since the team spans time zones.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

11.

A lender's approval model has 91% overall accuracy. It wrongly rejects 18% of qualified applicants over age 60 but 6% of other qualified applicants. A vendor built it in four months using 300 features and removed age as an input. What is the best conclusion?

- A. The model treats older applicants worse, likely through proxy features; investigate and mitigate using group-level metrics.
- B. It is fair because overall accuracy is 91%.
- C. It is fair because age was removed as an input.
- D. Four months was too short to build a fair model.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12. [TRANSFER — reversed stem / different variable]

A team wants to check whether its hiring model treats demographic groups equally. Which evaluation step reveals that?

- A. Nothing; models trained on historical data are always fair.
- B. Remove sensitive attributes and stop there.
- C. Compare selection rates and error rates across groups.
- D. Report overall accuracy on the test set.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

A wildlife agency's image model already recognizes 200 species. Rangers want it to recognize a newly listed species, but they have only 3 labeled photos of it, ~~taken by a field lab with 40 staff in a mountain region~~. Which approach fits?

**Target word:** learning setting for 3 labeled examples **Decides:** Three labeled examples of the new class.

|   |                   |                                                        |
|---|-------------------|--------------------------------------------------------|
| A | Absolute language | Few-shot methods are built for exactly this case.      |
| B | Confusion set     | Zero-shot means no labeled examples of the class.      |
| C | Adjacent category | Three images cannot train a model from zero.           |
| D | <b>Correct</b>    | A handful of labeled examples is the few-shot setting. |

## 2. Answer B Concept 2

A robotics lab ~~funded by a five-year grant~~ trains a policy across hundreds of different grasping tasks so that the robot can learn to grasp a brand-new object in a few trials. What is this approach?

**Target word:** name the approach **Decides:** Training over many tasks to adapt fast to new ones.

|   |                   |                                                                                  |
|---|-------------------|----------------------------------------------------------------------------------|
| A | Dropped element   | Drops the many-task structure that defines the method.                           |
| B | <b>Correct</b>    | It learns how to learn across a distribution of tasks.                           |
| C | Confusion set     | Combines models; does not learn to adapt.                                        |
| D | Adjacent category | Reuses one source task; it does not train for fast adaptation across many tasks. |

## 3. Answer D Concept 3

A team stacks gradient-boosted trees, a random forest, and logistic regression. It trains the meta-model on predictions the base models made on the same rows they were trained on. ~~Everything runs on a 30-node Kubernetes cluster~~. What is the problem?

**Target word:** flaw in the stacking setup **Decides:** Meta-model sees in-sample predictions.

|   |                    |                                                                     |
|---|--------------------|---------------------------------------------------------------------|
| A | Absolute language  | A leaky stack can do worse than its best base model.                |
| B | Camouflage         | Compute does not decide the leakage.                                |
| C | Inverted operation | Diversity is what makes ensembles work.                             |
| D | <b>Correct</b>     | Out-of-fold predictions show how base models behave on unseen data. |

## 4. Answer D Concept 4

A warehouse robot, ~~made by a firm operating in six countries with an 8-hour battery~~, will receive a reward of +1 at each of the next three steps, and then the episode ends. The discount factor is 0.9. What is the discounted return from now?

**Target word:** discounted return **Decides:**  $1 + 0.9 + 0.81$ .

|   |                   |                                                          |
|---|-------------------|----------------------------------------------------------|
| A | Dropped element   | Ignores the discount factor.                             |
| B | Camouflage        | Battery life does not enter the return.                  |
| C | Adjacent category | $0.9 + 0.81 + 0.729$ discounts the immediate reward too. |
| D | <b>Correct</b>    | $1 + 0.9(1) + 0.81(1) = 2.71$ .                          |

## 5. Answer C Concept 4 · transfer

An agent will receive one reward of 100 exactly two steps from now and nothing before. Its discounted value now is 64. What is the discount factor  $\gamma$ ?

**Target word:** solve for  $\gamma$  **Decides:**  $100\gamma^2 = 64$ .

|   |                    |                                                   |
|---|--------------------|---------------------------------------------------|
| A | Confusion set      | $1 - 0.64$ is the lost share, not $\gamma$ .      |
| B | Dropped element    | Forgets that two steps means $\gamma$ is squared. |
| C | <b>Correct</b>     | $\gamma^2 = 0.64$ , so $\gamma = 0.8$ .           |
| D | Inverted operation | $1 \div 0.8$ ; a discount factor cannot exceed 1. |

## 6. Answer C Concept 5

A recommendation agent ~~that serves 3 million users~~ uses  $\epsilon$ -greedy with  $\epsilon = 0.2$  across 4 possible actions; its random choice can land on any action, including the best one. What is the probability it picks the current best action?

**Target word:** probability of exploiting **Decides:**  $(1 - 0.2) + 0.2 \div 4$ .

|   |                    |                                                              |
|---|--------------------|--------------------------------------------------------------|
| A | Adjacent category  | Uniform random choice over 4 actions, ignoring $\epsilon$ .  |
| B | Inverted operation | That is the exploration rate.                                |
| C | <b>Correct</b>     | $0.8 + 0.05 = 0.85$ .                                        |
| D | Dropped element    | Leaves out the chance that a random pick is the best action. |

## 7. Answer B Concept 5 · transfer

An  $\epsilon$ -greedy agent with 5 actions picks its best-known action 92% of the time. What is  $\epsilon$ ?

**Target word:** solve for  $\epsilon$  **Decides:**  $(1 - \epsilon) + \epsilon \div 5 = 0.92$ .

|   |                    |                                                           |
|---|--------------------|-----------------------------------------------------------|
| A | Inverted operation | $0.08 \times 5$ multiplies instead of dividing by 0.8.    |
| B | <b>Correct</b>     | $1 - 0.8\epsilon = 0.92$ , so $\epsilon = 0.10$ .         |
| C | Dropped element    | $1 - 0.92$ ignores random picks that hit the best action. |
| D | Wrong stem number  | The exploit rate given in the stem.                       |

## 8. Answer A Concept 6

A fraud detector ~~used by 22 regional banks~~ flags 90% of fraudulent transactions and 5% of legitimate ones. Fraud makes up 1% of transactions. ~~The vendor recently won an industry award.~~ If a transaction is flagged, about how likely is it to be fraud?

**Target word:**  $P(\text{fraud} | \text{flag})$  **Decides:**  $0.009 \div (0.009 + 0.0495)$ .

|   |                    |                                                                                 |
|---|--------------------|---------------------------------------------------------------------------------|
| A | <b>Correct</b>     | $0.009 \div 0.0585 \approx 0.154$ .                                             |
| B | Wrong stem number  | The base rate before any evidence.                                              |
| C | Confusion set      | That is $P(\text{flag}   \text{fraud})$ , not $P(\text{fraud}   \text{flag})$ . |
| D | Inverted operation | $1 - 0.05$ is the legitimate pass-through rate.                                 |

## 9. Answer B Concept 6 · transfer

A screening model with 99% sensitivity and 99% specificity is applied to a condition that affects 1 in 10,000 people. A person screens positive. Which statement is correct?

**Target word:** reverse: interpret a positive under a tiny base rate **Decides:** About 1 true positive vs. about 100 false positives per 10,000.

|          |                   |                                                                |
|----------|-------------------|----------------------------------------------------------------|
| A        | Absolute language | It can still screen well when positives get follow-up testing. |
| <b>B</b> | <b>Correct</b>    | $0.99 \div (0.99 + 99.99) \approx 0.98\%$ .                    |
| C        | Confusion set     | Confuses sensitivity with the posterior.                       |
| D        | Dropped element   | Drops the base rate and error rates.                           |

## 10. Answer A Concept 7

For on-device keyboard prediction, Model A has 94.1% accuracy, is 1.2 GB, and takes 400 ms per prediction; Model B has 93.6% accuracy, is 60 MB, and takes 15 ms. The requirement is under 50 ms on phones. ~~Both were built by a team working across two time zones.~~ Which should be deployed?

**Target word:** model choice under constraints **Decides:** Only B meets the 50 ms limit.

|          |                   |                                                         |
|----------|-------------------|---------------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | A hard constraint rules out A.                          |
| B        | Absolute language | No such threshold is stated.                            |
| C        | Dropped element   | Drops the latency requirement.                          |
| D        | Camouflage        | Team location does not decide an on-device requirement. |

## 11. Answer A Concept 8

A lender's approval model has 91% overall accuracy. It wrongly rejects 18% of qualified applicants over age 60 but 6% of other qualified applicants. ~~A vendor built it in four months using 300 features~~ and removed age as an input. What is the best conclusion?

**Target word:** fairness judgment **Decides:** Error rates differ by group.

|          |                   |                                                                                      |
|----------|-------------------|--------------------------------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | Unequal false rejection rates are a fairness problem regardless of overall accuracy. |
| B        | Adjacent category | Overall accuracy hides group differences.                                            |
| C        | Dropped element   | Other features can stand in for age.                                                 |
| D        | Camouflage        | Build time does not decide the finding.                                              |

## 12. Answer C Concept 8 · transfer

A team wants to check whether its hiring model treats demographic groups equally. Which evaluation step reveals that?

**Target word:** reverse: from goal to method **Decides:** Fairness is measured per group.

|          |                   |                                           |
|----------|-------------------|-------------------------------------------|
| A        | Absolute language | Historical data often carries past bias.  |
| B        | Dropped element   | Proxies remain, and nothing is measured.  |
| <b>C</b> | <b>Correct</b>    | Group-level metrics expose disparities.   |
| D        | Adjacent category | One number cannot show group differences. |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 9             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 2             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 6             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 5             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 5             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 5             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 4             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                         | Rule                                                                                                                                                        | Trap                                                                                 | Tell                                                                        |
|---|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1 | <b>Zero-shot vs. few-shot learning</b>          | Few-shot learning adapts from a handful of labeled examples of a new class; zero-shot uses none and relies on descriptions or shared knowledge.             | Calling a task with a few examples zero-shot, or retraining from scratch on them.    | The count of labeled examples for the new class.                            |
| 2 | <b>Meta-learning</b>                            | Meta-learning trains across many related tasks so the model learns how to adapt quickly to a new task.                                                      | Equating it with training one bigger model on one task.                              | “Hundreds of tasks” and “adapt in a few trials.”                            |
| 3 | <b>Stacking and ensemble diversity</b>          | A stacking meta-model must learn from out-of-fold predictions, and ensembles gain most from diverse base models.                                            | Training the meta-model on in-sample predictions or combining near-identical models. | A meta-model trained on predictions from the same rows the base models saw. |
| 4 | <b>Discounted return</b>                        | Return $G = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots$ ; the immediate reward is not discounted.                                                              | Summing rewards without discounting or discounting the first reward.                 | A reward sequence and a discount factor $\gamma$ .                          |
| 5 | <b>Exploration vs. exploitation</b>             | Under $\epsilon$ -greedy, the best-known action is chosen with probability $(1 - \epsilon) + \epsilon \div$ number of actions.                              | Forgetting that a random pick can also land on the best action.                      | An $\epsilon$ value and a count of actions.                                 |
| 6 | <b>Bayes’ rule and base rates</b>               | $P(A \mid \text{evidence}) = P(\text{evidence} \mid A)P(A) \div P(\text{evidence})$ ; a rare base rate keeps the posterior low even for accurate detectors. | Reading the detector’s hit rate as the chance an alert is real.                      | A prevalence, a detection rate, and a false-alarm rate.                     |
| 7 | <b>Efficiency and sustainability trade-offs</b> | Choose models against the full requirement — accuracy, latency, memory, energy — not accuracy alone.                                                        | Picking the most accurate model even when it breaks a hard constraint.               | Two models with close accuracy and very different size or latency.          |
| 8 | <b>Evaluating fairness</b>                      | Measure error and selection rates by group; overall accuracy can hide disparities, and dropping a sensitive attribute does not remove proxies.              | Declaring a model fair because of high overall accuracy or a removed attribute.      | Different error rates across groups.                                        |

**Course scope (public description):** Advanced AI for Computer Scientists synthesizes principles of artificial intelligence (AI) and machine learning (ML) to equip students with the skills needed to design sophisticated AI systems that address real-world problems. The course explores cutting-edge techniques such as meta-learning, zero-shot and few-shot learning, and advanced ensemble methods. The course also covers state-of-the-art deep-learning architectures, reinforcement-learning strategies, and probabilistic-reasoning models. Additionally, the course prepares students to critically evaluate AI systems for performance, efficiency, sustainability, and ethical considerations, ensuring a comprehensive understanding of advanced AI applications.